

ARIAN MOGHNI

Santa Cruz, CA — amoghni@ucsc.edu — Last updated April, 2026

EDUCATION

University of California, Santa Cruz

September 2023 — June 2027

Bachelor of Science (B.S.) in Physics (Astrophysics)

Overall GPA: 4.0

Minor in History of Consciousness (Philosophy, Critical Theory, and Political Theory)

Relevant Coursework: Radiative Processes (Graduate), Astrophysical Fluid Dynamics (Graduate),
Order-of-Magnitude Astrophysics, Classical Mechanics, Electricity & Magnetism, Modern Physics

RESEARCH EXPERIENCE

Red Geyser Galaxies: Rowland Summer Research Fellowship

June 2024 - Present

Johns Hopkins University

Baltimore, MD

- Top 5% of applicants; invited back for a second summer.
- Led an investigation on mechanisms of quenching in galaxies by studying the cool & warm interstellar gas (Na I D and Ca II) in 140 red geyser galaxies ($z \sim 0.04$) using spatially resolved SDSS MaNGA IFU data.
- Two first-author papers (one in prep.); Best undergraduate research poster at annual APS regional meeting.
- **1 of 32 researchers from over 2,900 attendees ($\sim 1\%$) invited to the 247th American Astronomical Society (AAS) Press Conference.**
- **Goldwater Scholarship**

Mentors: Prof. Timothy Heckman (Johns Hopkins University), Dr. Namrata Roy (JHU & ASU), Prof. Kevin Bundy (UC Santa Cruz), and Dr. Kyle Westfall (UC Observatories)

Quantifying Confidence in Spectral Features

January 2025 - June 2025

UC Santa Cruz

Santa Cruz, CA

- Helped mentor a team of 4 first-year astrophysics undergraduates in applying statistical tools to the Hubble Space Telescope (HST) and James Webb Space Telescope (JWST) spectral data.
- Developed and implemented Gaussian fitting routines and χ^2 tests to assess the statistical significance of emission and absorption features in noisy spectra; applied these methods to datasets from HST and JWST.

Primary Mentor: Dr. Yayaati Chachan (UC Santa Cruz)

Modelling the Torus of Hot Jupiters: Where To Look For them?

January 2024 - June 2024

UC Santa Cruz

Santa Cruz, CA

- Created a model of how likely we can detect and discover the proposed torus around stars hosting Hot Jupiters with specific characteristics, such as stellar mass, stellar radius, temperature of gases, orbital distance, and ionizing flux, by simulating the observed spectra and the resulting absorption lines.
- The model can be used for narrowing down Hot Jupiter targets for observation and eventually lead to the discovery of the proposed torus around stars hosting Hot Jupiters.
- Presented findings at UC Santa Cruz ASTR 9 Research Symposium.

Mentors: Prof. Ruth Murray-Clay (UC Santa Cruz), Dr. Ethan Schreyer (Imperial College London)

Variable Stars: Science Internship Program (SIP)

June 2023 - August 2023

UC Santa Cruz

Santa Cruz, CA

- Top 9% of applicants; awarded full scholarship ($\sim \$9000$).
- Searched for new RR Lyrae stars in the Canada-France-Hawaii Telescope Legacy Survey (CFHTLS) data using the Lomb-Scargle periodogram and the Phase Distance Correlation method.
- Successfully tested a new heuristic formula for calculating the accumulated error in the estimated period of variable stars using the CFHTLS data. The formula can lead to more precise period measurements for variable stars and lead to the discovery of a greater number of these variable stars in the Vera C. Rubin Observatory Legacy Survey of Space and Time (LSST).

Mentor: Prof. Raja GuhaThakurta (UC Santa Cruz)

PUBLICATIONS

Moghni A., Heckman T. M., Roy N., Bundy K., Westfall K. B., & Rubin K. H. R., “Ca II Absorption as a Probe of Gas Flows in Red Geyser Galaxies”, in prep. (submit by August)

Moghni A., Roy N., Heckman T. M., Bundy K., Westfall K. B., & Rubin K. H. R. (2026). Galactic Rain: Cool Gas Inflows in Red Geyser Galaxies and Their Connection to AGN Activity and Interactions. *The Astrophysical Journal*, 1003, 31.

Moghni A., “The Spectrum: A Key to Understanding the Universe,” *Griffith Observer*, published February 2026.

SELECTED PRESS COVERAGE & MEDIA

United States

January 2026	<i>UC Santa Cruz News</i> <i>PHYS.org</i> (10 million monthly readers)	“How quiet galaxies stay quiet: cool gas feeds black holes in red geysers”
January 2026	<i>Universe Today</i> (5 million monthly views)	“Red geyser galaxies have plenty of star-forming gas but don’t form stars”

International

January 2026	<i>ICT Global</i> (Hungarian)	“Is a black hole the key to a great mystery in the universe?”
January 2026	<i>Fósil-Divulgación y Ciencia</i> (Spanish)	“How quiet galaxies stay quiet: cool gas feeds black holes in red geysers”

OBSERVING PROGRAMS

Giant Metrewave Radio Telescope (GMRT)

4 hours – (Co-I) *Revealing Aged Radio Plasma and AGN Feedback in “Red Geysers”: Old-and-Dead Galaxies with Low Luminosity AGNs and Large-Scale Biconical Outflows* GMRT Cycle 49 (2025)

Certified Observer, Nickel Telescope at Lick Observatory (2024–2025): Led 3 nights of observation for the Transients Team at UC Santa Cruz; PI: Professor Ryan Foley

AWARDS, GRANTS, AND SCHOLARSHIPS

International

Joan and Arnold Seidel *Griffith Observer* Science Writing Contest, 4th Prize (March 2025): The only undergraduate winner among international university faculty and professional authors honored in the contest. My article “The Spectrum: A Key to Understanding the Universe” was selected as one of the “best articles in astronomy, astrophysics, and space science, ... stimulating the flow of information between scientists, science writers, and the public.” My article was published in the *Griffith Observer* magazine in February 2026.

National

Barry M. Goldwater Scholarship (March 2026): Awarded to undergraduate students who show exceptional promise of becoming the Nation’s next generation of research leaders.

Iranian Scholarship Foundation Award (2023 - Present): \$40,000 total - given to the brightest and most promising Iranian Americans who may need financial support; requires at least a 3.5 GPA and more than 100 hours of community service each year.

APS Far West Section Best Undergraduate Research Poster (October 2025): Recognized by the American Physical Society “for an excellent poster presentation and outstanding achievement in undergraduate research” at the annual regional meeting (California, Hawaii, Nevada).

Vivien Thomas Scholars Initiative Summer Fellow (2025, 2024): \$14,000 total - the program and its fellowship is “dedicated to nurturing, mentoring and connecting exceptional diverse talent...to STEM graduate education and future leadership in STEM careers.”

SBB Research Group Foundation STEM Scholarship (August 2024): \$2,500 - Supports students who create significant value for society through their pursuit of higher learning in Science, Technology, Engineering, and Mathematics (STEM). Based on a clear plan to apply STEM innovations for a positive world impact.

Denes I. Bardos Award (July 2024): \$1,000 - Selection is based on exceptional achievement in science and community service, commitment to innovation, academic excellence, and significant resilience in overcoming challenges as an immigrant.

Science Internship Program (SIP) Full Scholarship (June 2023): Received 100% scholarship (equivalent to \$9,140) to cover the costs of the program and to complete my research on periodic variable stars.

University

Elmer A. Fridley Award (May 2026): \$2,000 - “Awarded annually alternating between an outstanding undergraduate and graduate student pursuing a degree in the physical sciences” (Physics, Earth & Planetary Sciences, Chemistry & Biochemistry, and Mathematics).

Koret Undergraduate Research Scholarship (November 2025): \$3,000 - Provides funding for undergraduate research projects in collaboration with faculty (red geyser galaxies research).

Ron Ruby Award (June 2025): \$4,000 - “Intended to reward the most promising young physicists...and students who have demonstrated potential for leadership in promoting cross-cultural understanding and/or those who have an outstanding record of service dedicated toward helping educationally disadvantaged students.”

John R. Lewis College Student Project Funds Grant (May 2025): \$575 - Travel grant for the 247th American Astronomical Society (AAS) meeting in Phoenix, AZ.

John R. Lewis College Student Project Funds Grant (November 2023): \$750 - Travel grant for the 243rd American Astronomical Society (AAS) meeting in New Orleans, LA.

Community

Bay East Foundation Scholarship (November 2023): \$2,500 - “In recognition of outstanding academic achievement and extracurricular contributions.”

Juanita Haugen Memorial Scholarship (April 2023): \$1,000 + awards from the Mayor of the city - given to 5 graduating students from three high schools in Pleasanton, CA for displaying a strong record of civic engagement & demonstrating Community of Character traits: compassion, honesty, respect, responsibility, integrity, and self-discipline.

PRESENTATIONS ($\triangle \equiv$ *Oral*, $\square \equiv$ *Poster*, $\star \equiv$ *Invited*)

Red Geyser Galaxies

$\square$ **UC Santa Cruz Admitted Students Day Research Exhibition (April 2026; Santa Cruz, CA):** Galactic Rain: Cool Gas Inflows in Red Geyser Galaxies

$\star$ **247th American Astronomical Society Press Conference (January 2026; Phoenix, AZ):** Galactic Rain: Cool Gas Inflows in Red Geyser Galaxies

$\square$ **247th American Astronomical Society iPoster (January 2026; Phoenix, AZ):** Galactic Rain: Cool Gas Inflows in Red Geyser Galaxies and their Connection to AGN Activity and Interactions

$\square$ **UC Santa Cruz Postdoc Symposium (November 2025; Santa Cruz, CA):** Galactic Rain in Red Geyser Galaxies: Links to AGN Activity and Galaxy Interactions

$\triangle$ **UC Santa Cruz No Jargon Talk (October 2025; Santa Cruz, CA):** Galactic Rain and the Death of Galaxies

- **American Physical Society Far West Section Meeting (October 2025; Santa Cruz, CA):** Galactic Rain in Red Geyser Galaxies: Links to AGN Activity and Galaxy Interactions
- △ **Leadership Alliance National Symposium (July 2025; Virtual):** The Distribution, Kinematics, and Origins of Cool Gas in Red Geyser Galaxies
- △ **Foothill High School Cosmos Club (September 2024; Pleasanton, CA):** Galaxies on Diet: The Origins of the Cool Gases of Red Geyser Galaxies
- **Stanford PIE Summer Workshop (September 2024; Palo Alto, CA):** Galaxies on Diet: The Origins of the Cool Gases of Red Geyser Galaxies
- **Johns Hopkins University Summer STEM Symposium (July 2024; Baltimore, MD):** Galaxies on Diet: The Origins of the Cool Gases of Red Geyser Galaxies
- **Leadership Alliance National Symposium (July 2024; Hartford, CT):** Galaxies on Diet: The Origins of the Cool Gases of Red Geyser Galaxies

RR Lyrae Variable Stars

- △ **Rotaract Research Symposium (June 2024; Virtual):** Dependence of the Width of the Periodogram Peak of Variable Stars on Observational and Light Curve Parameters
- △ **UC Santa Cruz No Jargon Talk (May 2024; Santa Cruz, CA):** A More Accurate Period for Variable Stars
- △ **Society of Physics Students Research Slam Presentation (February 2024; Santa Cruz, CA):** Dependence of the Width of the Periodogram Peak of Variable Stars on Observational and Light Curve Parameters
- **243rd American Astronomical Society Meeting (January 2024; New Orleans, LA):** Dependence of the Width of the Periodogram Peak of Variable Stars on Observational and Light Curve Parameters
- △ **Science Internship Program Presentation (August 2023; Santa Cruz, CA):** Searching for New RR Lyrae Stars and A New Path Toward More Accurate Period Measurements

Hot Jupiter Exoplanets

- **UC Santa Cruz ASTR 9 Research Symposium (June 2024; Santa Cruz, CA):** Modelling Parameters of Potential Tori of Hot Jupiters

MENTORING & TEACHING

Program Lead & Mentor: UC Santa Cruz Physics Mentoring Program September 2024 - Present
<https://spsclub.sites.ucsc.edu/physics-mentoring-program/>

- Program Lead: Connected 96 mentors and mentees to help undergraduates interested in physics and astrophysics get support for graduate school applications, summer REUs, handling classes, exploring different paths, or anything else they need support with. Led and hosted physics networking events.
- Mentor a first-year astrophysics undergraduate in research skill development, summer research program applications, and physics problem-solving.

Volunteer Tutor: Learn To Be Organization January 2024 - Present
<https://www.learntobe.org>

- Volunteer instructor of AP Calculus AB, AP Physics 1, Honors Algebra 2, and Geometry for high school and middle school students from underserved communities.

Mentor: Young Voyagers Mentoring Program August 2023 - Present

- Supporting high school students from underserved communities in physics and astronomy. Provide guidance on college applications, scholarships, and research opportunities.
- Mentored three students: one received a full-ride to the University of Pennsylvania, another accepted to Case Western as an international student, and one is a current high school senior.
- Led a college application workshop at Harbor High School in Santa Cruz, CA.

Research Peer Mentor: UC Santa Cruz Intro to Research Class

January 2025 - June 2025

- Research peer mentor for 4 first-year undergraduates. Assisted students with data analysis, computational tools, writing reports, and creating presentations from their findings.

Grader: PHYSICS 5C (Intro Electricity & Magnetism)

November 2024 - December 2024

- Graded problem sets, midterms, and the final + proctored for 129 undergraduates.

OTHER LEADERSHIP & OUTREACH ACTIVITIES

President: UC Santa Cruz Society of Physics Students (SPS)

September 2023 - Present

<https://www.instagram.com/ucscsps/>

- Selected outreach/leadership roles: Led the university-wide research showcase connecting undergraduates with the physics and astronomy faculty; hosted graduate school & REU workshops; raised over \$5,500; and led physics outreach events for community college and middle school students.
- Previous roles: treasurer & officer at large

Manager and Writer: NASA_Farsi_Ir (Instagram)

February 2022 - Present

https://www.instagram.com/nasa_farsi_ir/

- Write daily posts (1,300+) on new research findings and astronomy/physics basics for the general public in Farsi/Persian to help break language barriers.
- **500,000+** post views on Instagram and LinkedIn combined last year (same posts). Held live international meetings in the summer and discussed physics of stars (from birth to death), history of the solar system, calculating distances in astronomy, and future space missions for the general public.

Co-Fundraising Chair: Rotary Club of Santa Cruz

November 2023 - June 2025

<http://www.santacruzrotary.com>

- Planned and raised money for an economically disadvantaged family of four in Santa Cruz

OTHER INTERESTS

Long-distance Running: Most recent longest run was a half-marathon

Youth Soccer Referee: Referee up to U-19 for Elite Clubs National League (considered to be one of the most competitive youth soccer platforms in the country)